

Trigonometric Equations — Solutions

Warm-up

Question 1:

$$\begin{aligned}\sin x &= \frac{1}{2} \\ x &= \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6} \\ x &= \pi - \frac{\pi}{6} = \frac{5\pi}{6} \\ x &= \frac{\pi}{6}, \frac{5\pi}{6}\end{aligned}$$

Question 2:

$$\begin{aligned}R &= \sqrt{1^2 + 1^2} = \sqrt{2} \\ \tan \alpha &= \frac{1}{1} = 1 \implies \alpha = \frac{\pi}{4} \\ \sin x + \cos x &= \sqrt{2} \sin\left(x + \frac{\pi}{4}\right)\end{aligned}$$

Question 3:

$$\begin{aligned}f(x) &= 5 \cos\left(x - \frac{\pi}{6}\right) \\ \text{Range of } \cos\left(x - \frac{\pi}{6}\right) &= [-1, 1] \\ \text{Range of } f(x) &= [-5, 5] \\ \text{Max} &= 5, \text{Min} = -5\end{aligned}$$

Question 4:

$$\begin{aligned}\tan \alpha &= \sqrt{3} \\ \alpha &= \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}\end{aligned}$$

Question 5:

$$\begin{aligned}R \cos(x + \alpha) &= R(\cos x \cos \alpha - \sin x \sin \alpha) \\ &= R \cos \alpha \cos x - R \sin \alpha \sin x\end{aligned}$$

Question 6:

$$\begin{aligned}2 \cos \theta + 1 &= 0 \implies \cos \theta = -\frac{1}{2} \\ \theta &= \pi - \frac{\pi}{3} = \frac{2\pi}{3} \\ \theta &= -\left(\pi - \frac{\pi}{3}\right) = -\frac{2\pi}{3} \\ \theta &= \pm \frac{2\pi}{3}\end{aligned}$$

Question 7a:

$$R = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$

Question 7b:

$$R = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

Question 7c:

$$R = \sqrt{(\sqrt{5})^2 + 2^2} = \sqrt{5 + 4} = 3$$

Question 8:

$$R = \sqrt{1^2 + (\sqrt{3})^2} = 2$$
$$\tan \alpha = \frac{\sqrt{3}}{1} = \sqrt{3} \implies \alpha = 60^\circ$$

Question 9:

$$\sin 2x = \frac{1}{2}, \quad u = 2x$$
$$\sin u = \frac{1}{2} \implies u = \frac{\pi}{6}, \frac{5\pi}{6}$$
$$2x = \frac{\pi}{6} \implies x = \frac{\pi}{12}$$
$$2x = \frac{5\pi}{6} \implies x = \frac{5\pi}{12}$$

Question 10:

$$R = \sqrt{5^2 + (-12)^2} = 13$$
$$\tan \alpha = \frac{12}{5} \implies \alpha = \tan^{-1}(2.4)$$
$$5 \cos \theta - 12 \sin \theta = 13 \cos(\theta + \tan^{-1}(2.4))$$

Question 11:

$$\cos\left(x - \frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$
$$x - \frac{\pi}{4} = \pm \frac{\pi}{4}$$
$$x = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$$
$$x = -\frac{\pi}{4} + \frac{\pi}{4} = 0, 2\pi$$

Skill-building

Question 1:

$$R = \sqrt{5^2 + 12^2} = 13$$
$$\tan \alpha = \frac{12}{5} \implies \alpha = \tan^{-1}(2.4)$$
$$5 \sin x + 12 \cos x = 13 \sin(x + \tan^{-1}(2.4))$$

Question 2:

$$\begin{aligned}R &= \sqrt{(\sqrt{2})^2 + (-\sqrt{2})^2} = 2 \\2 \sin \left(x - \frac{\pi}{4}\right) &= 1 \implies \sin \left(x - \frac{\pi}{4}\right) = \frac{1}{2} \\x - \frac{\pi}{4} &= \frac{\pi}{6} \implies x = \frac{5\pi}{12} \\x - \frac{\pi}{4} &= \frac{5\pi}{6} \implies x = \frac{13\pi}{12}\end{aligned}$$

Question 3:

$$\begin{aligned}\cos 2\theta &= \frac{1}{2} \\2\theta &= \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3} \\\theta &= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}\end{aligned}$$

Question 4:

$$\begin{aligned}R &= \sqrt{2^2 + (-1)^2} = \sqrt{5} \\\tan \alpha &= \frac{1}{2} \implies \alpha = \tan^{-1}(0.5) \\2 \cos x - \sin x &= \sqrt{5} \cos(x + \tan^{-1}(0.5))\end{aligned}$$

Question 5:

$$\begin{aligned}2 \sin^2 \theta + \sin \theta - 1 &= 0 \\(2 \sin \theta - 1)(\sin \theta + 1) &= 0 \\\sin \theta &= \frac{1}{2} \implies \theta = \frac{\pi}{6}, \frac{5\pi}{6} \\\sin \theta &= -1 \implies \theta = \frac{3\pi}{2}\end{aligned}$$

Question 6:

$$\begin{aligned}R &= \sqrt{4^2 + (-3)^2} = 5 \\\tan \alpha &= \frac{3}{4} \implies \alpha = \tan^{-1}(0.75) \\4 \sin \theta - 3 \cos \theta &= 5 \sin(\theta - \tan^{-1}(0.75))\end{aligned}$$

Question 7:

$$\begin{aligned}\sin 2x &= \cos x \implies 2 \sin x \cos x - \cos x = 0 \\\cos x(2 \sin x - 1) &= 0 \\\cos x &= 0 \implies x = \frac{\pi}{2}, \frac{3\pi}{2} \\\sin x &= \frac{1}{2} \implies x = \frac{\pi}{6}, \frac{5\pi}{6}\end{aligned}$$

Question 8:

$$\begin{aligned}\sin x + \sqrt{3} \cos x &= 0 \\\tan x &= -\sqrt{3} \\x &= \pi - \frac{\pi}{3} = \frac{2\pi}{3}\end{aligned}$$

Question 9:

$$\tan^2 x = \frac{1}{3} \implies \tan x = \pm \frac{1}{\sqrt{3}}$$
$$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

Question 10:

$$\cos x - \sin x = \sqrt{2} \cos \left(x + \frac{\pi}{4} \right)$$
$$\sqrt{2} \cos \left(x + \frac{\pi}{4} \right) = \frac{1}{\sqrt{2}} \implies \cos \left(x + \frac{\pi}{4} \right) = \frac{1}{2}$$
$$x + \frac{\pi}{4} = \frac{\pi}{3} \implies x = \frac{\pi}{12}$$
$$x + \frac{\pi}{4} = \frac{5\pi}{3} \implies x = \frac{17\pi}{12}$$

Easier Exam Questions

Question 1:

$$R = \sqrt{(\sqrt{3})^2 + (-3)^2} = \sqrt{12} = 2\sqrt{3}$$
$$\tan A = \frac{3}{\sqrt{3}} = \sqrt{3} \implies A = \frac{\pi}{3}$$
$$\sqrt{3} \cos x - 3 \sin x = 2\sqrt{3} \cos \left(x + \frac{\pi}{3} \right)$$

Question 2:

$$R = \sqrt{3^2 + (-\sqrt{3})^2} = 2\sqrt{3}$$
$$2\sqrt{3} \sin \left(x - \frac{\pi}{6} \right) = \sqrt{3} \implies \sin \left(x - \frac{\pi}{6} \right) = \frac{1}{2}$$
$$x - \frac{\pi}{6} = \frac{\pi}{6} \implies x = \frac{\pi}{3}$$
$$x - \frac{\pi}{6} = \frac{5\pi}{6} \implies x = \pi$$

Question 3(i):

$$R = \sqrt{(\sqrt{12})^2 + 2^2} = 4$$
$$\tan \alpha = \frac{2}{\sqrt{12}} = \frac{1}{\sqrt{3}} \implies \alpha = \frac{\pi}{6}$$
$$\sqrt{12} \sin x + 2 \cos x = 4 \sin \left(x + \frac{\pi}{6} \right)$$

Question 3(ii):

$$4 \sin \left(x + \frac{\pi}{6} \right) = 2 \implies \sin \left(x + \frac{\pi}{6} \right) = \frac{1}{2}$$
$$x + \frac{\pi}{6} = \frac{\pi}{6} \implies x = 0$$
$$x + \frac{\pi}{6} = \frac{5\pi}{6} \implies x = \frac{2\pi}{3}$$

Question 4:

$$\begin{aligned}R &= \sqrt{(\sqrt{3})^2 + 1^2} = 2 \\ \tan \alpha &= \frac{1}{\sqrt{3}} \implies \alpha = \frac{\pi}{6} \\ \sqrt{3} \sin x + \cos x &= 2 \sin \left(x + \frac{\pi}{6} \right)\end{aligned}$$

Question 5:

$$\begin{aligned}3 \sin \theta &= 2\sqrt{3} \sin \theta \cos \theta \\ \sin \theta (3 - 2\sqrt{3} \cos \theta) &= 0 \\ \sin \theta = 0 &\implies \theta = 0, \pm\pi \\ \cos \theta = \frac{\sqrt{3}}{2} &\implies \theta = \pm \frac{\pi}{6}\end{aligned}$$

Question 6(i):

$$\begin{aligned}R &= \sqrt{1^2 + (-\sqrt{3})^2} = 2 \\ \tan \alpha &= \sqrt{3} \implies \alpha = \frac{\pi}{3} \\ \cos x - \sqrt{3} \sin x &= 2 \cos \left(x + \frac{\pi}{3} \right)\end{aligned}$$

Question 6(ii):

$$\begin{aligned}2 \cos \left(x + \frac{\pi}{3} \right) &= 1 \implies \cos \left(x + \frac{\pi}{3} \right) = \frac{1}{2} \\ x + \frac{\pi}{3} &= \frac{\pi}{3} \implies x = 0 \\ x + \frac{\pi}{3} &= \frac{5\pi}{3} \implies x = \frac{4\pi}{3}\end{aligned}$$

Question 7(i):

$$\begin{aligned}R &= \sqrt{3^2 + 2^2} = \sqrt{13} \\ \tan \alpha &= \frac{2}{3} \implies \alpha = \tan^{-1} \left(\frac{2}{3} \right) \\ 3 \cos x + 2 \sin x &= \sqrt{13} \cos(x - \alpha)\end{aligned}$$

Question 7(ii):

$$\begin{aligned}\sqrt{13} \cos(x - \alpha) &= \sqrt{13} \implies \cos(x - \alpha) = 1 \\ x - \alpha &= 0 \implies x = \alpha \approx 33.69^\circ\end{aligned}$$

Question 8:

$$\begin{aligned}2 \sin \left(\theta - \frac{\pi}{6} \right) &= 1 \implies \sin \left(\theta - \frac{\pi}{6} \right) = \frac{1}{2} \\ \theta - \frac{\pi}{6} &= \frac{\pi}{6} \implies \theta = \frac{\pi}{3} \\ \theta - \frac{\pi}{6} &= \frac{5\pi}{6} \implies \theta = \pi\end{aligned}$$

Question 9(i):

$$R = 2, \alpha = \frac{\pi}{3}$$
$$\cos x - \sqrt{3} \sin x = 2 \cos \left(x + \frac{\pi}{3} \right)$$

Question 9(ii):

$$2 \cos \left(x + \frac{\pi}{3} \right) = -2 \implies \cos \left(x + \frac{\pi}{3} \right) = -1$$
$$x + \frac{\pi}{3} = \pi \implies x = \frac{2\pi}{3}$$

Question 10(i):

$$R = 2, \alpha = \frac{\pi}{3}$$
$$\sin x + \sqrt{3} \cos x = 2 \sin \left(x + \frac{\pi}{3} \right)$$

Question 10(ii):

$$2 \sin \left(x + \frac{\pi}{3} \right) = 1 \implies \sin \left(x + \frac{\pi}{3} \right) = \frac{1}{2}$$
$$x + \frac{\pi}{3} = \frac{\pi}{6} \implies x = \frac{11\pi}{6}$$
$$x + \frac{\pi}{3} = \frac{5\pi}{6} \implies x = \frac{\pi}{2}$$

Question 11(i):

$$R = 2, \alpha = \frac{\pi}{3}$$
$$\cos x - \sqrt{3} \sin x = 2 \cos \left(x + \frac{\pi}{3} \right)$$

Question 11(ii):

$$2 \cos \left(x + \frac{\pi}{3} \right) = 2 \implies \cos \left(x + \frac{\pi}{3} \right) = 1$$
$$x + \frac{\pi}{3} = 2\pi \implies x = \frac{5\pi}{3}$$

Question 12(i):

$$R = 2, \alpha = \frac{\pi}{6}$$
$$\sqrt{3} \cos x - \sin x = 2 \cos \left(x + \frac{\pi}{6} \right)$$

Question 12(ii):

$$2 \cos \left(x + \frac{\pi}{6} \right) = 1 \implies \cos \left(x + \frac{\pi}{6} \right) = \frac{1}{2}$$
$$x + \frac{\pi}{6} = \frac{\pi}{3} \implies x = \frac{\pi}{6}$$
$$x + \frac{\pi}{6} = \frac{5\pi}{3} \implies x = \frac{3\pi}{2}$$

Harder Exam Questions

Question 1:

$$y = \cos x + \sin x = \sqrt{2} \cos\left(x - \frac{\pi}{4}\right)$$

Question 2:

$$\begin{aligned} 2 \cos^2 \theta - 1 &= \cos^2 \theta + \cos \theta + 1 \\ \cos^2 \theta - \cos \theta - 2 &= 0 \\ (\cos \theta - 2)(\cos \theta + 1) &= 0 \\ \cos \theta &= -1 \implies \theta = \pi \end{aligned}$$

Question 3(i):

$$\begin{aligned} A &= \sqrt{4^2 + 8^2} = 4\sqrt{5} \\ b &= \tan^{-1}(2) \\ I(t) &= 4\sqrt{5} \sin(0.02t + 1.107) \end{aligned}$$

Question 3(ii):

$$\begin{aligned} 0.02t + 1.107 &= \frac{\pi}{2} \approx 1.571 \\ 0.02t &= 0.464 \implies t = 23.2 \end{aligned}$$

Question 4:

$$\begin{aligned} \sqrt{2}[2 \cos x \cos(\pi/4)] &= 1 \\ 2 \cos x &= 1 \implies \cos x = \frac{1}{2} \\ x &= \pm \frac{\pi}{3} \end{aligned}$$

Question 5:

$$\begin{aligned} \sin x + \cos x &= \frac{\sqrt{6}}{2} \\ \sqrt{2} \sin(x + 45^\circ) &= \frac{\sqrt{6}}{2} \implies \sin(x + 45^\circ) = \frac{\sqrt{3}}{2} \\ x + 45^\circ &= 60^\circ, 120^\circ \\ x &= 15^\circ, 75^\circ \end{aligned}$$

Question 6:

$$\begin{aligned} 41 \sin(x + 1.35) &= 6 \implies \sin(x + 1.35) = 0.146 \\ x + 1.35 &= 2.995 \implies x = 1.645 \end{aligned}$$

Question 7:

$$\begin{aligned} \sqrt{2} \sin(\theta + \pi/4) &= -1 \implies \sin(\theta + \pi/4) = -\frac{1}{\sqrt{2}} \\ \theta + \pi/4 &= 5\pi/4, 7\pi/4 \\ \theta &= \pi, 3\pi/2 \end{aligned}$$

Question 8(i):

$$\tan 2x + \tan x = \frac{\sin 2x \cos x + \cos 2x \sin x}{\cos 2x \cos x} = \frac{\sin 3x}{\cos 2x \cos x}$$

Question 8(ii):

$$\begin{aligned}\sin 3x = 0 &\implies 3x = 0, \pi \\ x &= 0, \frac{\pi}{3}\end{aligned}$$

Question 9(i):

$$\begin{aligned}d(x) &= |\sqrt{2} \cos(x + \pi/4)| \\ d &= \sqrt{2}\end{aligned}$$

Question 9(ii):

$$\begin{aligned}\cos(x + \pi/4) &= -1 \implies x + \pi/4 = \pi \\ x &= \frac{3\pi}{4}\end{aligned}$$